IRRIGATION AND DRAINAGE

ASSINMENT 1

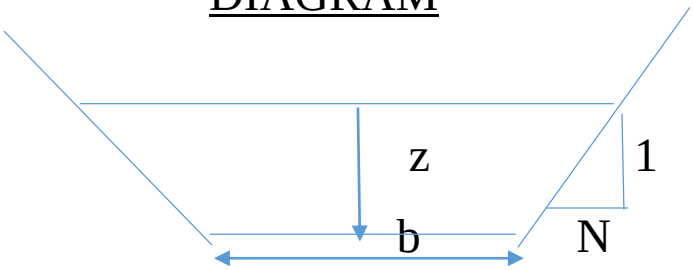
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QUESTION: 1

Design an economic trapezoidal section which carries a discharge of 1m/s with a bed slope of 1/1000 with Manning's coefficient of 0.02.assume a freeboard of 20%.taking N=2

SOLUTION

DIAGRAM



For an economic Trapezoidal section,

$$b = 2z \times ((N^2 + 1)^{\frac{1}{2}} - N)$$

$$P=2z(N^2 + 1)^{\frac{1}{2}} + b$$

$$A=(Nz + b) \times z$$

But

N=2, therefore

$$b = 2z((2^2 + 1)^{\frac{1}{2}} - 2)$$

$$b = 2z \times ((5^{\frac{1}{2}}) - 2))$$

$$b = 0.4721z$$

$$P = 2z(2^2 + 1)^{\frac{1}{2}} + 0.4721z$$

$$P = 4.472z + 0.4721z$$

$$P = 4.944z$$

$$A = (2z + 0.4721z) \times z$$

$$A = 2.4721z^2$$

$$R = \frac{A}{P} = \frac{2.4721z^2}{4.944z}$$

$$R = 0.5000z$$

$$Q = \frac{A}{n} \times (R)^{\frac{2}{3}} \times (S)^{\frac{1}{2}}$$

$$1 = \frac{2.4721z^2}{0.02} \times (0.500z)^{\frac{2}{3}} \times (\frac{1}{1000})^{\frac{1}{2}}$$

$$z = 0.7133m$$

FROM: $b = 2z \times ((N^2 + 1)^{\frac{1}{2}} - N)$

And

$z = 0.7133m$ and $N = 2$

Then $b = 2 \times 0.7133 \times ((2^2 + 1)^{\frac{1}{2}} - 2)$

$$b = 0.3368m$$

$$R = 0.5000z$$

$$R = 0.500 \times 0.7133$$

$$R = 0.35665m$$

$$A = 2.4721z^2$$

$$A = 2.4721 \times 0.7133$$

$$A = 1.76335m^2$$

$$P = 4.944z$$

$$P = 4.944 \times 0.7133$$

P =3.5266m

In the design of an economic trapezoidal section the parameters required are: z= 0.7133m and b= 0.3368m